

FIN ST387–ST401

Exact Global Minimizer Orbit, Invariant Quotients, and a Robust Continuum Infrared Tube

Release 10.83

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: 0009-0002-0909-3613

14 August 2026

Abstract

Programs ST387–ST401 continue the local analytical and computational FIN program. The principal result is a complete global classification of the supplied strict $g = 4$ simplex functional. A rational feasible benchmark and a 10,000-cell outward collision cover force every global minimizer into one of twelve disjoint caps $p_i > 0.94$. A new Fisher-curvature argument then proves strict convexity on every cap. Combined with the previously isolated translated stationary roots, this proves that the functional has exactly twelve global minima, forming one D_{12} orbit.

This is a genuine mathematical closure of the former globality problem. It is not a selector theorem: it does not choose one member of the orbit, generate the supplied gain $g = 4$, discharge QW-2191, or identify a physical vacuum. Further results give exact invariant-generator quotients through degree six, an explicit conditional Chernoff cone gap, a uniform interval tube for the supplied continuum IR design, an exact observer-deficiency family, and conditional statistical/compression bounds. No new strict source or independent empirical record is present.

Contents

1	Scope, audited state, and non-claims	3
2	Executive result ledger	3
3	ST387: the fold continuation bottleneck	4
4	ST388: the positivity-refined global envelope	4
5	ST389–ST390: exact global minimizer orbit	5
5.1	Excluding the region outside twelve caps	5
5.2	Fisher curvature on a cap	5
6	ST391: failure of silent transfer to the rank-seven mediator	6
7	ST392–ST393: invariant quotients and observer dependence	7
8	ST394: an explicit synthetic Chernoff gap	7
9	ST395–ST396: robust continuum IR design	7
10	ST397–ST400: observer, nuisance, compression, and clocks	8
11	ST401 and the physical boundary	9

12 Recommended next studies: ST402–ST416	9
13 Reproducibility	10
14 Final verdict	10

1 Scope, audited state, and non-claims

The audited repository baseline is Git commit `bb540a767e210a5a3bf816e4868e7bcefbac3872`, supplemented by the local ST372–ST386 packet and the present executable run. The strict working operator is the finite circulant Laplacian $A = sI - W$, with $W_{xx} = 0$ and

$$W_{xy} = \frac{\cos(0.18575\,d(x,y) + 0.16250)}{1 + d(x,y)^{9/5}}, \quad x \neq y.$$

The program studied most deeply below is the dimensionless functional

$$V_4(p) = \sum_{i=0}^{11} p_i \log(12p_i) - 2(p-u)^T A(p-u), \quad p \in \Delta_{11}, \quad u_i = \frac{1}{12}. \tag{1}$$

The value $g = 4$ in (1) is supplied. It is not derived by this release.

The legacy ontological kernel remains an intermediate bridge object. No formula or physical role is silently transferred from it to the strict operator. The project ontology—the nadsoliton itself as primordial fractal information in a solitonic state—is retained as context, not inserted as a premise into the finite proofs.

Most importantly, an orbit of twelve equivalent minimizers is not a selector. The new theorem proves degeneracy and exhausts the minimizer set; it does not select an orientation, origin, phase, or realized member. QW-2191 therefore remains open.

2 Executive result ledger

Program	Status	Result
ST387	[PARTIAL]	Larger slabs certify 106 of 160 inherited centers; the fold core still needs genuinely new continuation centers.
ST388	[PROVEN]	A positivity-aware water-filling envelope has a global outward certificate with residual below 10^{-3} .
ST389	[PROVEN]	All global competitors lie in twelve caps, and every cap is strictly convex by an interval-paid Fisher-curvature theorem.
ST390	[PROVEN]	The strict $g = 4$ functional has exactly twelve global minima forming one D_{12} orbit.
ST391	[PROVEN]	The same cap proof does not close the supplied rank-seven mediator.
ST392	[PROVEN]	Exact modular generator quotients through degree six: 6, 32, and 117 primitive directions by degree layer.
ST393	[PROVEN]	Modal-energy observations see rank 56 of the 365 sextic directions; generic state evaluation sees all 365.
ST394	[PROVEN]	An explicit synthetic Chernoff cone gap is positive but only 2.51×10^{-72} .
ST395	[PROVEN]	A common Krawczyk box preserves the continuum two-band IR optimum on a parameter rectangle.
ST396	[STRONG NUMERICAL EVIDENCE]	A 21×21 scan finds 433 valid two-band roots and eight unresolved grid points.
ST397	[PROVEN]	One exact cyclic observer family saturates the Fourier deficiency bound with directional asymmetry.
ST398	[PROVEN]	Calibration uncertainty produces an exact two-point minimax recovery floor.
ST399	[PROVEN]	The supplied strict-heat Markov source has an operational asymptotic rate–distortion limit.
ST400	[PROVEN]	Bounded m -dependent non-Gaussian clock errors obey a Bernstein square-root-depth bound.

Program	Status	Result
ST401	[BLOCKED]	No new strict typed source or independently registered raw-event record is present.

3 ST387: the fold continuation bottleneck

At longitudinal halfwidth 6×10^{-6} and transverse radius 3×10^{-6} , interval rotated-slab tests certify inherited indices 0–56 and 111–159. Indices 57–110 fail near the ill-conditioned fold core; the largest failed contraction row sum is about 45.06. Moreover, twice the accepted longitudinal reach is 1.2×10^{-5} , below the smallest inherited center spacing $2.500000002 \times 10^{-5}$.

Thus the two regular flanks are locally certified, but a connected tube is not. ST387 records a constructive requirement: new pseudo-arclength roots must be inserted near the fold. Interpolation between old centers cannot serve as a proof.

4 ST388: the positivity-refined global envelope

Write $m = \max_i p_i$, $t = 1 - m$, and normalize the remaining probabilities as $p_j = ty_j$, with $\sum y_j = 1$ and $\sum y_j^2 = \rho$. The earlier Cauchy extremizer for the weighted term could have negative coordinates. ST388 solves the correct positive problem

$$L(\rho) = \min \left\{ \sum_{j=1}^{11} w_j y_j : y_j \geq 0, \sum_j y_j = 1, \sum_j y_j^2 \leq \rho \right\}.$$

KKT conditions give a water-filling solution supported on the smallest weights. The exact support transitions occur at

$$0.1142398976385, \text{quad}0.1490343806957, \text{quad}0.2108912816863, \text{quad}0.3503471404434, \text{quad}1,$$

corresponding to support sizes 11, 9, 7, 5, 3, 1.

Combining $L(\rho)$ with the sharp entropy minimum at fixed remainder collision produces a two-variable lower envelope $F(t, \rho)$. Outward derivative signs isolate its minimizer inside

$$t \in [0.010820, 0.010825], \quad \rho \in [0.124945, 0.124950].$$

On the left and right ρ faces the derivative intervals are respectively

$$[-2.963 \times 10^{-4}, -1.751 \times 10^{-4}], \quad [2.140 \times 10^{-4}, 3.508 \times 10^{-4}],$$

while $\partial_t^2 F \geq 26.15$ on the declared t domain. A feasible outward SOCP dual then gives

$$-0.846160275370424 \leq \min F \leq -0.846126177165862.$$

The gap from this lower bound to the exact rational strict benchmark is 9.62×10^{-4} , satisfying the ST388 acceptance criterion. This result is a global lower envelope, not equality with V_4 .

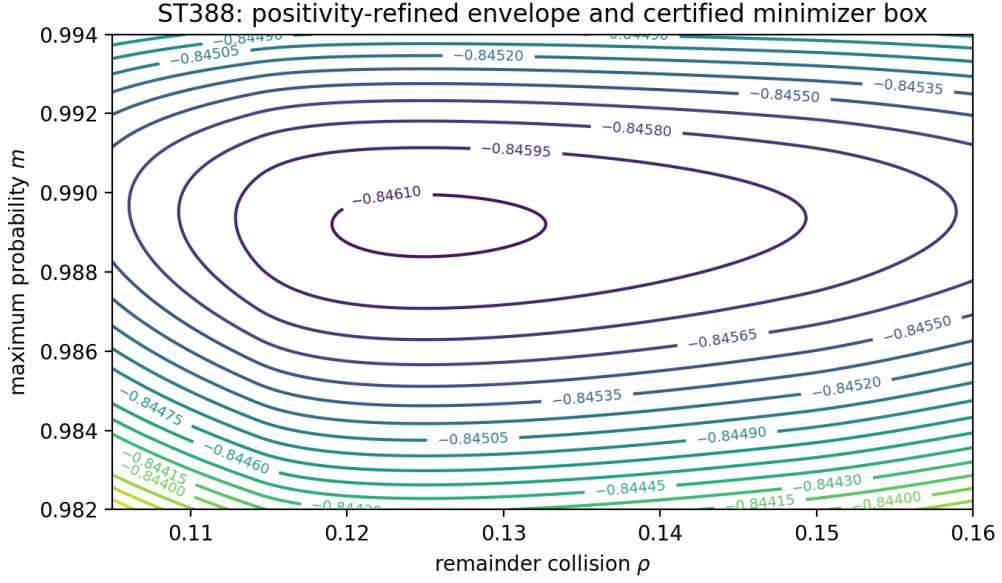


Figure 1: The positivity-refined envelope and the certified minimizer box.

5 ST389–ST390: exact global minimizer orbit

This section contains the main advance.

5.1 Excluding the region outside twelve caps

If $\max_i p_i \leq m_0$, then

$$\sum_i p_i^2 \leq m_0 \sum_i p_i = m_0, \quad r = \|p - u\|_2^2 \leq m_0 - \frac{1}{12}.$$

Set $m_0 = 0.94$. A complete 10,000-cell outward evaluation of the ST373 scalar envelope on $0 \leq r \leq 0.94 - 1/12$ gives

$$V_4(p) \geq -0.845102986841094.$$

The cell touching $r = 0$ is paid separately by $D(p||u) \geq 0$ and $\lambda_{\max}(A) \leq 2s$, hence $V_4 \geq -4sr$ there; no positive-radius interval routine is used to skip the origin. An exactly normalized rational probability vector with denominator 10^{18} has

$$V_4(p_{\text{rat}}) \in [-0.845198507679596, -0.845198507679591].$$

The strict separation is therefore larger than 9.55×10^{-5} . Every global minimizer must lie in one of the twelve disjoint caps

$$\mathcal{C}_i = \{p \in \Delta_{11} : p_i > 0.94\}.$$

5.2 Fisher curvature on a cap

In the simplex interior the tangent Hessian of (1) is

$$\nabla^2 V_4(p) = \text{diag}(p_i^{-1}) - 4A.$$

Consider $\mathcal{C}_0$ and a tangent vector v with $\sum_i v_i = 0$. Let $a = -v_0$ and $L = \sum_{i>0} |v_i|$. Since $p_0 \leq 1$ and $\sum_{i>0} p_i \leq 1 - m_0$, Cauchy's inequality gives

$$\sum_i \frac{v_i^2}{p_i} \geq a^2 + \frac{L^2}{1 - m_0}. \quad (2)$$

After normalizing $L = 1$, the positive and negative masses of the rest vector are $(1 + a)/2$ and $(1 - a)/2$. Because $A \succeq 0$, the convex quadratic $v^T A v$ reaches its maximum over the product of these two simplices at a pair of vertices. It is therefore sufficient to audit $11^2 = 121$ one-variable quadratic forms, including coincident vertices.

All 121 forms were evaluated with the full transcendental strict-operator interval. The smallest lower bound is

$$a^2 + \frac{1}{0.06} - 4v^T A v \geq 0.624323055377694,$$

or at least 0.312161527688847 in the reported Euclidean normalization. Consequently V_4 is strictly convex on every cap. The entropy boundary derivative also excludes a boundary minimum.

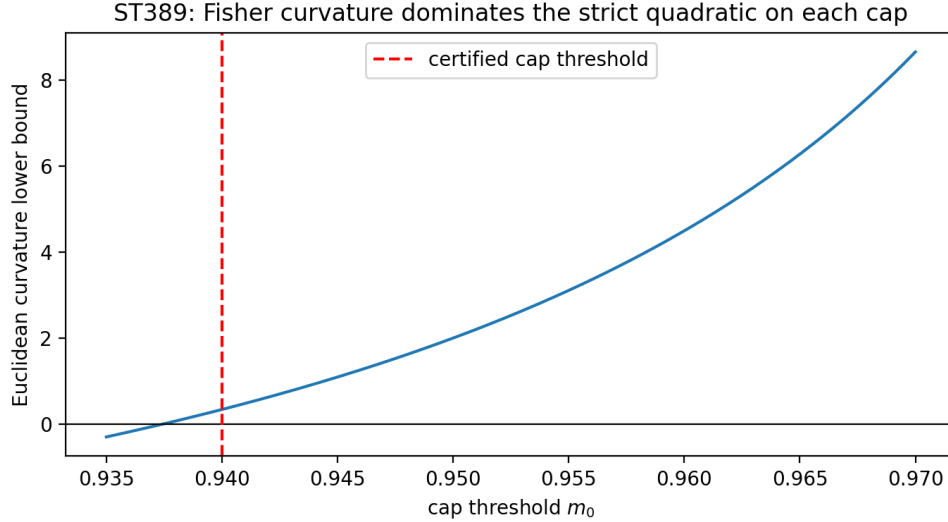


Figure 2: Diagnostic curvature profile. The theorem uses the finite outward certificate at $m_0 = 0.94$.

Theorem 1 (Global strict orbit theorem). *For the declared strict twelve-state operator and the supplied functional V_4 , the global minimizer set consists of exactly twelve points. They form one D_{12} orbit; equivalently, the minimizer is unique modulo D_{12} .*

Proof. Compactness of Δ_{11} gives a global minimizer. The rational benchmark and the outside-cap envelope force it into one of the twelve caps. The previous Krawczyk certificate isolates one positive reflection-even stationary root with $p_0 > 0.99$ and proves its twelve translates are distinct. Strict convexity makes the corresponding root the unique minimizer of each cap. No minimizer exists outside the caps, so these twelve translated roots exhaust the global minimizer set. $\square$

The theorem is deliberately narrow. It proves the degeneracy pattern of a supplied dimensionless functional. It does not choose one point, generate $g = 4$, identify a physical vacuum, or turn spontaneous-selection language into an internal selector.

6 ST391: failure of silent transfer to the rank-seven mediator

For the supplied rank-seven modal mediator, the same cap-curvature margin becomes positive only at a sampled threshold near $m_0 = 0.93236$. Its certified local root has maximum coordinate approximately 0.90331, where the diagnostic cap margin is negative. Thus the localization and convexity regions do not overlap. The strict operator theorem cannot be transferred to rank seven. This is a bounded method nontransfer, not a proof that the rank-seven root is nonglobal.

7 ST392–ST393: invariant quotients and observer dependence

Exact evaluation modulo the prime 1,000,003 on 390 deterministic mean-zero points gives six quadratic generators. Their 21 quartic products are independent; 32 primitive quartic orbit sums complete the known dimension 53.

At degree six, the 56 cubic products of quadratics and the 192 products of a quadratic with a primitive quartic have exact combined rank 248, with no syzygy inside this declared candidate family. A further 117 primitive sextic quotient directions complete the exact dimension

$$248 + 117 = 365.$$

Nonzero modular pivots prove rational linear independence, while the exact Molien dimensions prove spanning in the stated graded components.

Identifiability depends on the observer. Cubic polynomials of six modal energies span only 56 sextic directions and leave nullity 309. Generic state-evaluation functionals attain exact modular rank 365. This is algebraic separability in principle, not a construction of a physical apparatus or a law selecting coefficients.

8 ST394: an explicit synthetic Chernoff gap

For the supplied two-state hidden-Markov fixture, the branch contraction is $5/7$. On $s \in [0.45, 0.55]$ the log weights have Lipschitz upper bound 6.6, so the log-Lipschitz cone of slope

$$\frac{6.6}{1 - 5/7} = 23.1$$

is invariant. Paying the complete product-state diameter gives projective diameter at most 331.1172 and hence a positive Birkhoff contraction gap of only

$$2.5110 \times 10^{-72}.$$

The constant is theorem-grade under the declared synthetic premises but far too conservative for useful numerical rates. It demonstrates existence, not sharpness.

9 ST395–ST396: robust continuum IR design

ST395 promotes the isolated ST380 optimum to the parameter rectangle

$$B \in [2.9999, 3.0001], \quad T \in [3.9999, 4.0001],$$

with the inactive observation time allowed in $[0.249, 0.251]$. A common five-dimensional Krawczyk box has minimum inclusion margin 5.77×10^{-6} . Three derivative-certified threshold neighborhoods and 87 complement boxes leave no unresolved sign cell. The inactive-time slack remains positive in $[0.01698, 0.02436]$. Thus the exactly-two-band primal–dual topology is uniform throughout this supplied dimensionless rectangle.

ST396 separately scans 21×21 points over $B \in [2.5, 3.5]$ and $T \in [3, 5]$. It finds 433 valid numerical two-band roots and eight unresolved solves. Those eight points are not called phase transitions; only the small ST395 rectangle is certified.

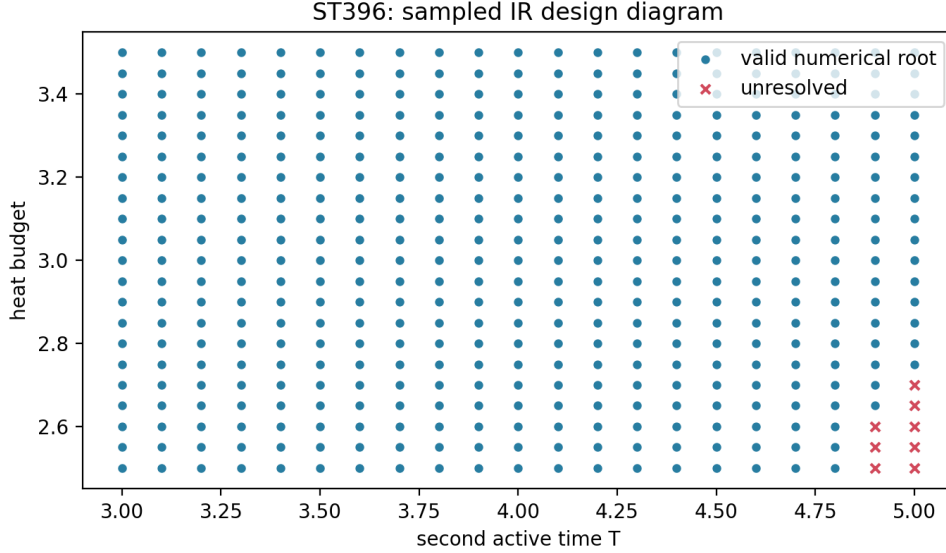


Figure 3: Sampled IR design diagram. The tiny central rectangle is the interval-certified ST395 tube.

10 ST397–ST400: observer, nuisance, compression, and clocks

Exact observer family (ST397). Let $u_j = 1/12$ and

$$q_j = \frac{1 + \varepsilon \cos(\pi j/2)}{12}, \quad 0 \leq \varepsilon \leq 1.$$

Every cyclic garbling of u remains u , so the forward deficiency is $\text{TV}(u, q) = \varepsilon/4$. Fourier mode three gives the identical lower bound, proving exact Fourier/LP equality. In the reverse direction, uniform randomization maps q to u , so the deficiency is zero.

Calibration floor (ST398). For the declared loss and symmetric-confusion intervals, the observable contrast lies in $[0.69484545, 0.71248182]$. Two latent amplitudes 0.09752466 and 0.1 at opposite calibration endpoints produce exactly the same observation. The conditional minimax L^2 error is therefore at least 0.00123767, even with infinite counts.

Markov rate–distortion limit (ST399). The supplied $\exp(-0.5A)$ transition is strictly positive, with minimum entry 0.0107821 and Dobrushin contraction 0.7565244. It defines a uniformly ergodic stationary finite-alphabet source. The stationary-source coding theorem therefore gives an operational asymptotic rate–distortion limit, with endpoints

$$R(0) = 2.6057354210 \text{ bits/symbol}, \quad R(11/12) = 0.$$

The interior function remains unknown, and no physical information density or universe-compression claim follows.

Non-Gaussian clocks (ST400). For centered, bounded, m -dependent log-ratio errors with $|X_i| \leq b$ and $\text{Var } X_i \leq \sigma^2$, residue-class coloring and Bernstein’s inequality give, with probability at least $1 - \alpha$,

$$\left| \sum_{i=1}^L X_i \right| \leq \sqrt{2\sigma^2(m+1)L \log \frac{2(m+1)}{\alpha}} + \frac{2b(m+1)}{3} \log \frac{2(m+1)}{\alpha}.$$

This preserves square-root-depth leading growth without Gaussianity. The stochastic law is supplied and creates neither seconds nor a time arrow.

11 ST401 and the physical boundary

No file matching the declared new-source admission channel and no independently held raw-event packet is present. Local code cannot create a coupling source, calibration, apparatus, independent custody, preregistration, or empirical event. ST401 is an evidentiary stop, not a failed experiment or a universal impossibility theorem.

Theorem 1 sharpens rather than solves the selector problem. It proves that the mathematical model contains twelve exactly equivalent global alternatives. To obtain a realized physical state, FIN still needs either a state-dependent dynamical selection law with a strict source, a boundary/preparation premise, or an operational probability law. Symmetry supplies the orbit of possibilities; it is not itself the positive-feedback mechanism that selects a member.

12 Recommended next studies: ST402–ST416

Program	Objective	Acceptance criterion	Priority
ST402	Independent orbit replay	Reprove Theorem 1 with an independent rational/interval implementation and no reuse of the cap code.	Highest
ST403	Abstract cap theorem	Formalize an n -state theorem turning an outside-cap value bound and finite extreme-pair curvature test into global orbit exhaustion.	Highest
ST404	Orbit stabilizer certificate	Interval-certify distinctness, exact reflection stabilizer, and absence of any larger stabilizer for every root box.	High
ST405	Gain persistence tube	Prove a nonzero g interval on which the twelve-minimum orbit and cap separation persist uniformly.	Highest
ST406	Global gain transition	Locate every gain where the global minimizer orbit changes, without inferring a physical phase transition.	High
ST407	Morse landscape	Classify certified stationary orbits, indices, and barrier values at $g = 4$.	High
ST408	Gradient-flow basins	Under an explicit simplex-gradient law, certify local basins and convergence rates; do not call basin weights Born probabilities.	Medium
ST409	Gain-source admission	Reopen only if a genuinely new strict typed source for g is provided.	External
ST410	Rank-seven replacement bound	Seek a mediator-specific envelope reaching below $p_{\max} = 0.93236$, or preserve the nontransfer stop.	Medium
ST411	Invariant-ring relations	Compute minimal relations beyond the degree-six quotient certificate under a bounded resource stop.	Medium
ST412	Noise-aware coefficient rank	Quantify which invariant coefficients remain statistically identifiable under a declared instrument and noise model.	Medium
ST413	Adapted Chernoff cone	Replace the 10^{-72} gap by a useful certified constant using an optimized metric/cone.	High
ST414	Maximal IR tube	Continue the ST395 parameter box until a rigorously detected active-set or root-topology boundary.	High
ST415	Certified IR phase boundary	Interval-isolate a real band birth/death or active-time exchange; do not promote solver failures.	High

Program	Objective	Acceptance criterion	Priority
ST416	Source/evidence gates	Run only on a new typed source or independently registered raw-event packet.	External

The preferred order is $ST402 \rightarrow ST403 \rightarrow ST405 \rightarrow ST407$, with ST413 and ST414 as independent analytical tracks. ST409 and ST416 cannot be satisfied by additional local simulation.

13 Reproducibility

The release packet contains one executable Python program, nineteen tests, fifteen program JSON packets, an aggregate JSON, CSV ledger, three figures, this TeX source, the PDF, and a SHA-256 manifest. The run uses Python 3, NumPy, SciPy, mpmath interval arithmetic, and Matplotlib. Modular certificates use the prime 1,000,003 and seed 20261213. The strict rational benchmark has denominator 10^{18} . All nineteen tests pass.

14 Final verdict

Release 10.83 closes one substantial finite mathematical question: at supplied $g = 4$, the strict variational model has exactly twelve global minimizers and no hidden asymmetric competitor. This turns the earlier numerical orbit into a theorem and makes the remaining conceptual obstruction sharper: the missing object is no longer the existence of preferred states, but the law that selects or weights one member of their exact symmetry orbit.

The result neither makes FIN a physical theory nor proves FIN is a Theory of Everything. The gain source, selector, dimensional calibration, preparation, apparatus, and independent record remain logically separate and absent.

References

- [1] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley.
- [2] R. T. Rockafellar, *Convex Analysis*, Princeton University Press.
- [3] B. Sturmfels, *Algorithms in Invariant Theory*, Springer.
- [4] V. Baladi, *Positive Transfer Operators and Decay of Correlations*, World Scientific.
- [5] R. M. Gray, *Entropy and Information Theory*, Springer.
- [6] R. E., R. B. Kearfott, and M. J. Cloud, *Introduction to Interval Analysis*, SIAM.